

# Helix Thready — Reused Submodule Map & Maturity

Table of Contents

1. How to read this map

2. Data & infrastructure

3. AI / LLM

4. Skills & processing

5. Messenger & assets

6. Security & auth

7. Frontend / design / mobile

8. Testing / QA / docs / process

9. Dependency & maturity diagram

10. Decoupling constraints [\$11.4.28]

| Rev | Date       | Author                      | Change                                                                                                                                                                                                                                                                                                                                                                                          |
|-----|------------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1   | 2026-07-21 | swarm (development)         | Initial map — every reused module, role, import path, maturity                                                                                                                                                                                                                                                                                                                                  |
| 2   | 2026-07-22 | swarm (development, pass 3) | Pass-3 source verification — <code>session_orchestrator</code> upgraded DESIGN-ONLY → <b>FOUNDATION</b> (real <code>claim</code> registry read at source); <code>LLM0rchestrator</code> <code>AgentPool</code> contract confirmed; <code>Agentic</code> confirmed to exist; herald <code>channels.Channel</code> <code>seam</code> pinned to <code>commons_messaging/channels/channel.go</code> |

This document maps **every** `vasic-digital` / `HelixDevelopment` / `milos85vasic` submodule Helix Thready reuses or extends, its role, its import path (where confirmed), and its **maturity** as recorded in the private gap register. It is the “in-house first” ledger: the decision matrix (final request \$0.2) chose an owned module for almost every concern, and this map records the honest build-readiness of each so no scaffold is mistaken for production.

**Maturity legend** `[gap register $0]` · `PRODUCTION` (build on directly) · `FOUNDATION` (real but early) · `SCAFFOLD` (skeleton/stub) · `DESIGN-ONLY` (no working code) · `BUILD-NEW` (does not exist — see `build-new-subsystems.md`). **Confidence:** `VERIFIED` (read at source) · `FLAGGED` (metadata/README only — re-verify before reliance).

## Table of Contents

---

- 1. How to read this map
- 2. Data & infrastructure
- 3. AI / LLM
- 4. Skills & processing
- 5. Messenger & assets
- 6. Security & auth
- 7. Frontend / design / mobile
- 8. Testing / QA / docs / process
- 9. Dependency & maturity diagram
- 10. Decoupling constraints `[$11.4.28]`

## 1. How to read this map

---

- **Role** — what Thready uses it for.
- **Import / repo** — the confirmed import path ( `digital.vasic.*` convention) or repo slug.
- **Maturity / Conf.** — from the gap register.
- **Thready action** — `WIRE` (config-inject as-is) · `EXTEND` · `HARDEN` (raise a scaffold) · `BUILD-NEW`. Cross-referenced to the `ATM-NNN` item(s) and `[GAP: ...]`.

## 2. Data & infrastructure

---

| Module     | Role                                                                                           | Import / repo                         | Maturity / Conf.                                 | Thready action                                 | Item / Gap                   |
|------------|------------------------------------------------------------------------------------------------|---------------------------------------|--------------------------------------------------|------------------------------------------------|------------------------------|
| database   | Relational store (SQLite dev / PostgreSQL prod; <code>migration.Runner()</code> )              | <code>digital.vasic.database</code>   | PRODUCTION / VERIFIED                            | WIRE + add partitioning/retention helpers      | ATM-003, ATM-043 · [GAP 3.2] |
| vectordb   | Semantic store (pgvector or cosine; Qdrant/ Pinecone/ Milvus behind <code>VectorStore</code> ) | <code>digital.vasic.vectordb</code>   | PRODUCTION (pgvector) / VERIFIED; others FLAGGED | WIRE pgvector; HARDEN Qdrant                   | ATM-004, ATM-042 · [GAP 3.1] |
| embeddings | Embedding generation (OpenAI-compatible providers)                                             | <code>digital.vasic.embeddings</code> | PRODUCTION / VERIFIED                            | EXTEND: add native llama.cpp/HelixLLM provider | ATM-041 · [GAP 2.7]          |
| rag        | Retrieval-augmented generation                                                                 | <code>digital.vasic.rag</code>        | PRODUCTION / VERIFIED                            | WIRE                                           | ATM-039                      |
| eventbus   | In-proc channel bus + <b>NATS JetStream</b>                                                    | <code>digital.vasic.eventbus</code>   | PRODUCTION / VERIFIED                            | WIRE (JetStream primary)                       | ATM-011                      |

| Module     | Role                                                      | Import / repo                         | Maturity / Conf.      | Thready action                                      | Item / Gap                   |
|------------|-----------------------------------------------------------|---------------------------------------|-----------------------|-----------------------------------------------------|------------------------------|
|            | durable adapter                                           |                                       |                       |                                                     |                              |
| background | Postgres task queue, DLQ, retry/back-off, worker pool     | <code>digital.vasic.background</code> | PRODUCTION / VERIFIED | WIRE (idempotent single-claim)                      | ATM-023, ATM-027             |
| messaging  | Kafka/Rabbit MQ for firehose streams                      | <code>digital.vasic.messaging</code>  | PRODUCTION / VERIFIED | WIRE (optional, high-throughput)                    | —                            |
| cache      | L1/L2 in-mem + Redis + Postgres; consistent hashing       | <code>digital.vasic.cache</code>      | PRODUCTION / VERIFIED | WIRE                                                | —                            |
| storage    | MinIO/S3 object store + local FS + signed URLs            | <code>digital.vasic.storage</code>    | PRODUCTION / VERIFIED | WIRE; validate MinIO signed-URL parity              | ATM-034, ATM-043 · [GAP 3.2] |
| filesystem | SMB/FTP/NFS/WebDAV/local; <code>OpenSeal</code> for Range | <code>digital.vasic.filesystem</code> | PRODUCTION / VERIFIED | EXTEND: <b>add HTTP(S) source</b> ; fix NFS listing | ATM-029 · [GAP 6.2]          |

| Module                         | Role                                                           | Import / repo                                | Maturity / Conf.      | Thready action                      | Item / Gap       |
|--------------------------------|----------------------------------------------------------------|----------------------------------------------|-----------------------|-------------------------------------|------------------|
| observability                  | OTel + Prometheus + logrus + ClickHouse + pkg/health           | digital.vasic.observeability                 | PRODUCTION / VERIFIED | WIRE                                | ATM-005          |
| containers                     | Rootless Podman orchestration ( pkg/ boot / compose / health ) | vasic-digital/containers                     | PRODUCTION / VERIFIED | WIRE (sole orchestration \$11.4.76) | ATM-002          |
| lets_encrypt                   | ACME TLS (HTTP-01/ DNS-01, atomic deploy-hook + rollback)      | vasic-digital/lets_encrypt                   | PRODUCTION / VERIFIED | WIRE per subdomain                  | ATM-006          |
| discovery / mdns / port_prefix | Service discovery + deterministic dynamic ports                | digital.vasic.discovery · mdns · port_prefix | PRODUCTION / VERIFIED | WIRE                                | ATM-007          |
| http3                          | quic-go/ http3 transport                                       | vasic-digital/http3                          | PRODUCTION / VERIFIED | WIRE (REST + Download Manager)      | ATM-020, ATM-028 |

| Module                      | Role                                              | Import / repo                                                                          | Maturity / Conf.      | Thready action | Item / Gap |
|-----------------------------|---------------------------------------------------|----------------------------------------------------------------------------------------|-----------------------|----------------|------------|
|                             | wrapper                                           |                                                                                        |                       |                |            |
| middleware /<br>ratelimiter | CORS/<br>request-id/<br>recover;<br>rate limiting | <code>digital.vasic.middleware</code> ·<br><code>digital.vasic.rate<br/>limiter</code> | PRODUCTION / VERIFIED | WIRE           | ATM-020    |

## 3. AI / LLM

---

| Module               | Role                                                                                            | Import / repo                                          | Maturity / Conf.                                                                                                                                      | Thready action                                                                                           | Item / Gap                                   |
|----------------------|-------------------------------------------------------------------------------------------------|--------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------|----------------------------------------------|
| HelixLLM             | Local llama.cpp serving (OpenAI+Anthropic APIs); embeddings endpoint                            | <code>HelixDevelopment/HelixLLM</code>                 | PRODUCTION <b>with traps</b> / VERIFIED                                                                                                               | HARDEN: enforce <code>llama</code> embedder, kill <code>HashEmbedder</code> default; remove 768 hardcode | ATM-040, ATM-041 · <b>[GAP 2.1]</b>          |
| LLMProvider          | 40+ provider adapters; retry/circuit-breaker/health                                             | <code>digital.vasic.llmprovider</code>                 | PRODUCTION / FLAGGED (adapters unaudited)                                                                                                             | WIRE + per-adapter contract tests                                                                        | ATM-064 · <b>[GAP 2.3]</b>                   |
| LLMOrchestrator      | Dev-time headless-agent pool orchestration ( <code>AgentPool</code> )                           | <code>vasic-digital/LLMOrchestrator (pkg/agent)</code> | PRODUCTION / <b>VERIFIED</b> ( <code>AgentPool</code> contract read at source: <code>Acquire(ctx, AgentRequirements) (Agent,error) / Release</code> ) | WIRE for dev fleet                                                                                       | ATM-067 ( <b>closed</b> ) · <b>[GAP 2.4]</b> |
| HelixAgent           | Ensemble/debate agent hub                                                                       | <code>dev.helix.agent</code>                           | FOUNDATION / FLAGGED (identity blur)                                                                                                                  | HARDEN: resolve identity split; pin only needed pkgs                                                     | ATM-063 · <b>[GAP 2.2]</b>                   |
| LLMsVerifier         | Model scoring/selection for the fallback chain                                                  | <code>digital.vasic.llmsverifier</code>                | PRODUCTION core / VERIFIED                                                                                                                            | HARDEN: reconcile <code>:7061 / :8080</code> ; stable scoring API                                        | ATM-065 · <b>[GAP 2.5]</b>                   |
| VisionEngine         | LLM-vision adapters; CV analyzer ( <code>StubAnalyzer</code> behind <code>-tags vision</code> ) | <code>digital.vasic.visionengine</code>                | FOUNDATION / VERIFIED ( <b>no OCR</b> )                                                                                                               | EXTEND: add <code>OCRProvider</code> seam                                                                | ATM-033 · <b>[GAP 2.6]</b>                   |
| Memory / HelixMemory | Semantic/cognitive memory                                                                       | <code>digital.vasic.memory</code> · HelixMemory        | PRODUCTION / VERIFIED                                                                                                                                 | DEFER or back <code>Memory</code> with <code>vectordb</code>                                             | — · <b>[GAP 2.8]</b>                         |
| token_optimizer      | Prompt token optimization pipeline                                                              | <code>digital.vasic.token_optimizer</code>             | <b>partial/WIP</b> / VERIFIED                                                                                                                         | HARDEN or scope-out                                                                                      | ATM-061 · <b>[GAP 2.9]</b>                   |
| TOON                 | Token-Oriented Object Notation encoder                                                          | <code>digital.vasic.TOON</code>                        | <b>SCAFFOLD (not implemented)</b> / VERIFIED                                                                                                          | HARDEN or scope-out; use toolkit <code>toon.mjs</code> meanwhile                                         | ATM-061 · <b>[GAP 2.9]</b>                   |

| Module                                                                                                 | Role                                                                         | Import / repo                                                                                                                    | Maturity / Conf.                                                                                               | Thready action                                                                                                               | Item / Gap          |
|--------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------|---------------------|
| session_orchestrator                                                                                   | Atomic track-claim registry (exactly-once claim + deadlock-free device-lock) | <code>vasic-digital/session_orchestrator (claim/, alias/, scheduler/, supervisor/)</code>                                        | DESIGN-ONLY<br><b>FOUNDATION / VERIFIED</b> (Pass 3 — real <code>Registry.TryClaim / Release</code> at source) | <b>BUILD WIRE + verify + integrate</b> the shipped registry (realizes \$11.4.176(A); reused by runtime single-claim ATM-023) | ATM-062 · [GAP 2.9] |
| Normalize / conversation / SkillRegistry / ToolSchema / MCP_Module / Agentic / Planning / AgentWrapper | Agent-support modules                                                        | <code>vasic-digital/Agentic confirmed</code> (“graph-based agentic workflow orchestration”); other seven paths still unconfirmed | Agentic VERIFIED; rest FLAGGED (docs-only)                                                                     | <b>Source-verify + pin</b> the residual seven before reliance                                                                | ATM-068 · [GAP 2.9] |

**Anti-bluff, do not misread.** HelixLLM is production **with a trap** — its default local embedder is a non-semantic `HashEmbedder` stub (VERIFIED at `HelixLLM/internal/knowledge/embedding_providers.go`: `NewEmbedder` maps `"hash"/"local"` **and any unrecognised provider name** to `NewHashEmbedder`); `TOON` has **no working code** (`ErrTOONEncodingNotImplemented`). **Correction (Pass 3):** `session_orchestrator` — previously listed here as “no working code” — **now ships a real exactly-once claim `Registry`** at source, so it is upgraded to FOUNDATION and is **no longer** cited as a no-code gap. The honest rule stands: a claim is only made when it has been read at source.

## 4. Skills & processing

| Module                     | Role                                                         | Import / repo                                 | Maturity / Conf.                                         | Thready action                                                         | Item / Gap                                   |
|----------------------------|--------------------------------------------------------------|-----------------------------------------------|----------------------------------------------------------|------------------------------------------------------------------------|----------------------------------------------|
| helix_skills               | Skill-Graph DAG (atomic→composite→umbrella); knowledge units | <a href="#">HelixDevelopment/helix_skills</a> | FOUNDATION/MVP / VERIFIED ( <b>no execution engine</b> ) | BUILD-NEW dispatch engine on top; standardize <a href="#">SKILL.md</a> | ATM-025, ATM-037, ATM-038 · <b>[GAP 4.1]</b> |
| SkillRegistry / MCP_Module | Runtime skill registration + tool exposure                   | (paths unconfirmed)                           | FLAGGED                                                  | Source-verify                                                          | ATM-068 · [GAP 2.9]                          |
| helix_proto                | Protobuf/ OpenAPI codegen pattern                            | <a href="#">HelixDevelopment/helix_proto</a>  | PRODUCTION / VERIFIED                                    | WIRE for SDK codegen                                                   | ATM-060                                      |

## 5. Messenger & assets

| Module             | Role                                                                                                              | Import / repo                                                                                                                                       | Maturity / Conf.                                                                                                                                                                                                              | Thready action                                                                                                                                                                                                                                                                 | Item / Gap                                            |
|--------------------|-------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------|
| herald             | Messenger<br>s/posts/<br>replies +<br>fan-out;<br><code>gotd/td</code><br>MTPROTO<br>in<br><code>qa Herald</code> | <code>vasic-digital/</code><br><code>herald</code> (seam:<br><code>commons_messaging</code><br><code>/channels/</code><br><code>channel.go</code> ) | FOUNDATION / VERIFIED<br><b>(MTPROTO trapped in QA;</b><br><b>Max stub</b> — no<br><code>channels/max</code> or<br><code>channels/mtpROTO</code> pkg,<br>only <code>docs/guides/</code><br><code>messagingers/MAX.md</code> ) | EXTEND: promote<br>MTPROTO reader; add<br>Max; ThreadReader —<br>all implement<br><code>channels.Channel</code><br>( <code>SendReplyGeneric</code> /<br><code>BotSelfIdentity</code> /<br><code>DownloadAttachmen</code><br><code>t</code> +<br><code>commons.Channel</code> ) | ATM-016,<br>ATM-017,<br>ATM-018 ·<br><b>[GAP 5.1]</b> |
| Catalogizer        | Multi-<br>protocol<br>asset store<br>(SQLCipher,<br>JWT+RBAC,<br>WS)                                              | <code>vasic-digital/</code><br><code>Catalogizer</code>                                                                                             | PRODUCTION / VERIFIED<br>(not decoupled)                                                                                                                                                                                      | BUILD-NEW: decouple<br>as Asset Service                                                                                                                                                                                                                                        | ATM-012 ·<br>[GAP 6.1]                                |
| Boba-Base          | Torrent<br>search/<br>download<br>(SSE +<br><code>POST</code> /<br><code>api/v1/</code><br><code>hooks</code> )   | <code>milos85vasic/</code><br><code>Boba-Base</code>                                                                                                | FOUNDATION / VERIFIED                                                                                                                                                                                                         | EXTEND: standardize<br>callback                                                                                                                                                                                                                                                | ATM-032 ·<br>[GAP 6.4]                                |
| YT-DLP<br>(MeTube) | Video/<br>streaming<br>download<br>(poll-only<br>API)                                                             | <code>milos85vasic/YT-</code><br><code>DLP</code>                                                                                                   | FOUNDATION / VERIFIED<br><b>(no outbound webhook)</b>                                                                                                                                                                         | EXTEND: add<br>completion webhook                                                                                                                                                                                                                                              | ATM-031 ·<br>[GAP 6.5]                                |
| streaming          | WebSoc<br>ket hub ( <b>not</b><br>media<br>byte<br>streaming)                                                     | <code>digital.vasic.str</code><br><code>eamg</code>                                                                                                 | PRODUCTION / VERIFIED                                                                                                                                                                                                         | WIRE for real-time<br>surface                                                                                                                                                                                                                                                  | ATM-021                                               |

## 6. Security & auth

| Module       | Role                                                                          | Import / repo                                        | Maturity / Conf.                            | Thready action                                           | Item / Gap                   |
|--------------|-------------------------------------------------------------------------------|------------------------------------------------------|---------------------------------------------|----------------------------------------------------------|------------------------------|
| security     | AES-256-GCM + Argon2 id; pkg/secur estor age / pii / polic y / heade rs; SSRF | <code>digital.vasic.secur</code><br><code>ity</code> | PRODUCTION / VERIFIED                       | WIRE; add searchable-sealed creds; wire scanner adapters | ATM-014, ATM-055 · [GAP 7.1] |
| auth         | JWT + API keys + OAuth 2 (default HMAC-SHA256)                                | <code>digital.vasic.auth</code>                      | PRODUCTION / VERIFIED                       | EXTEND: RS256/EdDSA + JWKS; RBAC into User Service       | ATM-013, ATM-010 · [GAP 7.2] |
| Security-KMP | Mobile secure storage                                                         | <code>vasic-digital/Security-KMP</code>              | <b>SCAFFOLD (in-memory stub)</b> / VERIFIED | HARDEN: real Keychain/KeyStore before mobile release     | ATM-015 · <b>[GAP 7.3]</b>   |
| Auth-KMP     | Mobile OAuth 2 flows                                                          | <code>vasic-digital/Auth-KMP</code>                  | FOUNDATION / VERIFIED                       | EXTEND: consume fixed Security-KMP                       | ATM-046 · [GAP 7.4]          |

## 7. Frontend / design / mobile

---

| Module                                                                                        | Role                                                     | Import / repo                               | Maturity / Conf.                                                         | Thready action                                         | Item / Gap          |
|-----------------------------------------------------------------------------------------------|----------------------------------------------------------|---------------------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------|---------------------|
| design_system                                                                                 | Web/CSS + Angular tokens / components (helix-green base) | <a href="#">vasic-digital/design_system</a> | FOUNDATION / VERIFIED                                                    | HARDEN + Thready brand theme; publish npm              | ATM-050 · [GAP 8.1] |
| open-design                                                                                   | Design-system source of truth (§11.4.162)                | <a href="#">nexu-io/open-design</a>         | PRODUCTION / VERIFIED                                                    | WIRE (source, never fork)                              | ATM-050             |
| helix_design                                                                                  | Non-web design arm (Flutter/Qt/CSS token pkgs)           | <a href="#">vasic-digital/helix_design</a>  | <b>SCAFFOLD</b> / VERIFIED                                               | HARDEN: implement per-platform pkgs                    | ATM-051 · [GAP 8.2] |
| helix_ui                                                                                      | Flutter UI (alt family)                                  | <a href="#">vasic-digital/helix_ui</a>      | <b>SCAFFOLD</b> / VERIFIED                                               | Only if Flutter chosen; cannot target HarmonyOS/Aurora | — · [GAP 8.3]       |
| KMP fleet (Auth/Security/Storage/Database / Concurrency/Formatters/Document/Config/RateLimit) | Shared mobile logic                                      | <a href="#">vasic-digital/*-KMP</a>         | <b>SCAFFOLD (no CI/publish; Database-KMP interfaces-only)</b> / VERIFIED | HARDEN: CI + Maven + SQLDelight                        | ATM-047 · [GAP 8.4] |

| Module                                   | Role                                        | Import / repo                                           | Maturity / Conf.                    | Thready action                         | Item / Gap          |
|------------------------------------------|---------------------------------------------|---------------------------------------------------------|-------------------------------------|----------------------------------------|---------------------|
| er/UI-Components-KMP)                    |                                             |                                                         |                                     |                                        |                     |
| helix_shims + Harmony OS/ Aurora clients | Native platform reach                       | <code>vasic-digital/helix_shims</code> + native clients | <b>SCAFFOLD</b> / VERIFIED/ FLAGGED | BUILD-NEW: ArkTS + Qt clients          | ATM-046 · [GAP 8.5] |
| helix_track_cli                          | TUI reference (Bubble Tea/ Cobra/ Lipgloss) | <code>vasic-digital/helix_track_cli</code>              | FOUNDATION / FLAGGED                | Confirm impl before adopting wholesale | ATM-049 · [GAP 8.6] |
| TS client libs (6) + UI-Components-React | React/ TS surfaces                          | (various)                                               | SCAFFOLD/FOUNDATION / FLAGGED       | Re-audit + CI                          | ATM-049 · [GAP 8.6] |

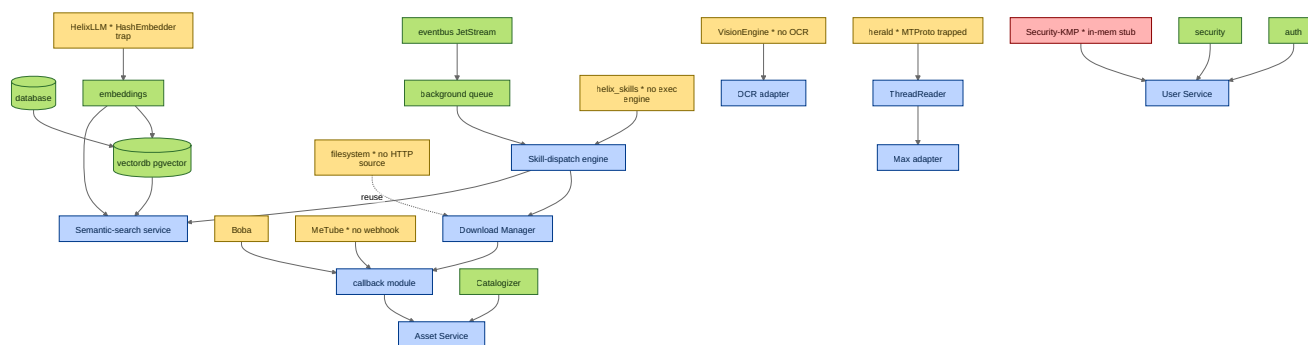
## 8. Testing / QA / docs / process

---

| Module                                                                          | Role                                                                        | Import / repo                                                                            | Maturity / Conf.                                               | Thready action                                      | Item / Gap                          |
|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------------------------------------------|-----------------------------------------------------|-------------------------------------|
| challenge<br>s                                                                  | Per-<br>feature<br>real-<br>use-<br>case<br>scenari<br>o bank               | <code>vasic-digital/<br/>challenges</code>                                               | PRODUCTION / VERIFIED                                          | WIRE: author Thready<br>scenario banks              | ATM-054 · [GAP<br>9.3]              |
| HelixQA /<br>helix_qa                                                           | AI-<br>driven<br>multi-<br>platfor<br>m QA<br>(runti<br>me<br>eviden<br>ce) | <code>HelixDevelopment/<br/>helix_qa</code> (+ <code>vasic-<br/>digital/HelixQA</code> ) | PRODUCTION / VERIFIED/<br>FLAGGED (mirror)                     | WIRE: YAML banks;<br>confirm canonical vs<br>mirror | ATM-054 · [GAP<br>9.1]              |
| Panoptic<br>/<br>VisualReg<br>ression /<br>ScreenDif<br>f /<br>ReplayBu<br>ffer | Visual<br>regress<br>ion<br>(LLM-<br>vision<br>+ pixel)                     | <code>vasic-digital/*</code>                                                             | Panoptic PRODUCTION; VR<br>family no-CI / VERIFIED/<br>FLAGGED | WIRE + add CI to VR<br>family                       | ATM-054 · [GAP<br>9.3]              |
| DocProce<br>ssor                                                                | Featur<br>e-map<br>→<br>covera<br>ge<br>trackin<br>g                        | <code>HelixDevelopment/<br/>DocProcessor</code>                                          | PRODUCTION / VERIFIED                                          | WIRE docs↔tests<br>coverage                         | ATM-054 · [GAP<br>9.4]              |
| docs_cha<br>in                                                                  | md↔H<br>TML/<br>PDF/<br>DOCX↔<br>SQLite<br>sync<br>(\$11.4.<br>65/106<br>)  | <code>vasic-digital/<br/>docs_chain</code>                                               | PRODUCTION (ph 1-5) /<br>VERIFIED                              | WIRE; provision<br>pandoc/weasyprint                | ATM-009,<br>ATM-069 · [GAP<br>10.1] |
| claude_t<br>oolkit                                                              | Multi-<br>accoun<br>t                                                       | <code>vasic-digital/<br/>claude_toolkit</code>                                           | PRODUCTION / VERIFIED                                          | WIRE for dev fleet<br>(alias-first, token opt)      | orchestration                       |

| Module                       | Role                                                                                | Import / repo                                                                                | Maturity / Conf.      | Thready action                    | Item / Gap |
|------------------------------|-------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|-----------------------|-----------------------------------|------------|
|                              | aliases<br>;<br>provid<br>er<br>aliases<br>; TOON<br>utils;<br>OpenC<br>ode<br>sync |                                                                                              |                       |                                   |            |
| HelixTran<br>slate +<br>i18n | Localiz<br>ation<br>(en/ru/<br>sr-Cyrl)                                             | <code>HelixDevelopment/<br/>helix_translate</code> ·<br><code>digital.vasic.i18<br/>n</code> | PRODUCTION / VERIFIED | WIRE                              | ATM-044    |
| codegrap<br>h                | Code<br>intellig<br>ence<br>(\$11.4.<br>78–80)                                      | <code>colbymchenry/<br/>codegraph</code>                                                     | PRODUCTION / VERIFIED | WIRE: index own-org<br>submodules | ATM-008    |

## 9. Dependency & maturity diagram



**Explanation (for readers/models that cannot see the diagram).** Colour encodes maturity: green = PRODUCTION, amber = FOUNDATION (real but early, with the specific trap noted inline), red = SCAFFOLD/stub, blue = BUILD-NEW. Reading the colours before the arrows is the point of the diagram — it shows at a glance where Thready builds on solid ground and where it must first harden or build.

The **data spine** is mostly green and wired directly: `database` → `vectordb` (pgvector) with `embeddings`. The one amber node in it is `HelixLLM`, which feeds `embeddings` but carries the `HashEmbedder`-default trap ( `ATM-040` ) — production code with a stub embedder behind it, which is why it is not drawn green. The **processing spine** then flows `eventbus` → `background` → the **new** (blue) Skill-dispatch engine, which reads the amber `helix_skills` Skill-Graph (knowledge/ordering only — no execution engine of its own).

From the dispatch engine the **asset/download river** runs: it drives the **new** Download Manager (reusing the amber `filesystem`, which lacks an HTTP source), which emits the **new** standardized callback that `Boba` and the amber `MeTube` (no webhook today) also emit, all landing in the **new** Asset Service decoupled from the green `Catalogizer`. Two more amber modules gain blue extensions on the side: `VisionEngine` (no OCR) gains a **new** OCR adapter, and `herald` (MTPROTO trapped in a QA harness, Max an empty stub) gains a **new** ThreadReader and Max adapter.

The **service backing** completes the picture: the green `auth` + `security` plus the **red** `Security-KMP` stub back the **new** User Service (the red node is why mobile token storage is a release blocker, `ATM-015`), and the dispatch engine together with the green `embeddings` / `vectordb` feed the **new** Semantic-search service. The invariant the diagram enforces: every blue node has a design plan in `build-new-subsystems.md`, and every amber/red trap carries a `[GAP]` and a hardening item — no node is relied on for more than its colour honestly permits.

Rendered PNG/SVG exported via Docs Chain (§11.4.65). Source: `diagrams/submodule-maturity.mmd`.

## 10. Decoupling constraints `[§11.4.28]`

VERIFIED at source (Constitution §11.4.28 / CONST-051): every owned submodule Thready reuses MUST be **project-not-aware**, fully reusable and **config-injected**, laid out at `<root>/<name>/` or `<root>/submodules/<name>/`. **Nested own-org submodule chains are forbidden** (depth-1 exception only for constitution-anchored reusable engines carrying `helix-deps.yaml`). Consequences for this map: (1) Thready never forks a module — it consumes it as a dependency and extends **upstream**; (2) each **BUILD-NEW** submodule (Asset Service, Download Manager, Max adapter, OCR adapter, User Service, callback module) gets its **own** repo under `vasic-digital / HelixDevelopment` with an `upstreams/` recipe, not a vendored copy inside `helix_thready`; (3) `ATM-059` audits every reused module for project-not-awareness before wiring.

Made with love ♥ by Helix Development.